

Unavoidable cycle-contraction minors of large 2-connected graphs

Wayne Ge James Oxley

October 6, 2024

1 Introduction

Skipped for now

2 Preliminary

For graph and matroid terminology that we do not define explicitly, we follow [4] and [6]. For graphs G and H , we write $H \leq_{cc} G$ and say that H is a *circuit-contraction minor* or a *cc-minor* of G if there is a sequence $G_0, G_1, \dots, G_k$ of graphs such that $(G_0, G_k) = (G, H)$ and, for all i in $[k]$, there is a cycle C_{i-1} of G_{i-1} such that $G_i = G_{i-1}/C_{i-1}$. Because a loop is a single-edge cycle, each contraction of a loop from a graph produces a cc-minor of the graph. Thus, from now on, we shall assume that each contraction of a cycle is accompanied by the contraction of all loops that one created. Loops aren't interesting to our result, so we assume all the graphs we are dealing with are loopless.

For any graph G , we say H is its *parallel extension* if H can be obtained from G by adding edges parallel to existing edges in G . We say a graph G is *k-connected* if $|V(G)| \geq k$ and for any $A \subset V(G)$ such that $|A| < k$, $G \setminus A$ is connected. A graph G is *k-edge-connected* if for any $B \subseteq E(G)$ such that $|B| < k$, $G \setminus B$ is connected. Note that the clique on 2 vertices, K_2 , is 2-connected but not 2-edge-connected. However, any nontrivial parallel extension of K_2 are both 2-connected and 2-edge-connected. Particularly, if a graph is a parallel extension of K_2 , we say it is a *dipole graph* D_n where $n = |E(D_n)|$. For any connected graph G , a *2-block* of G is a maximal 2-connected subgraph of G .

A *fan graph* F_n is a simple graph that is obtained from an n -vertex path $v_1 v_2 \dots v_n$ by joining each v_i to a new vertex u . We call each uv_i edge a *spoke*, and particularly each uv_1 or uv_n edge an *outer spoke*. For positive integers $t_1, t_2, \dots, t_n$, we obtained a *fan-type graph* $F_{(t_1, t_2, \dots, t_n)}$ by replacing the spokes uv_i of F_n by t_i parallel edges. In Figure 1, we show three different examples of fan-type graphs. Note that dipole graphs are considered to be fan-type graphs.

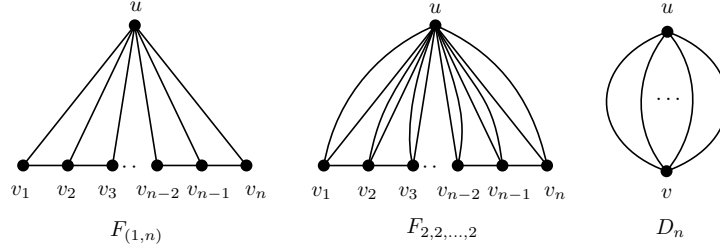


Figure 1: Three examples of fan-type graphs

A *direct-sum* (or *1-sum*) G of G_1, G_2 is obtained by identifying a vertex v_1 of G_1 and v_2 of G_2 . Note that any connected graph G can be viewed as direct-sum of 2-blocks of G . A *2-sum* G of G_1, G_2 is obtained by identifying a nonloop edge e_1 of G_1 with a nonloop edge e_2 of G_2 and deleting the identified edge. So G depends on the choice of e_1, e_2 , and how e_1, e_2 are identified (how their ends are paired).

An *r-template* is a 2-connected graph G obtained from an r -vertex path P_r using the following operations:

- i. Label each vertex of P_r by K_4, K_3 , or D_3 (we call these vertex labels *building blocks*);
- ii. For each building block pick one edge for each of its adjacent building block. Moreover, if a building block labeled by K_4 is adjacent to two other blocks, then we always pick two non-adjacent edges;
- iii. Apply 2-sums on specified edges.

Figure 2 shows the building blocks for templates, and Figure 3 shows the constructing process of a sample 6-template.

The following are some special examples of templates:

1. A fan graph F_n is a $(2n - 3)$ -template in which the building blocks alternate between K_3 and D_3 , beginning with K_3 .

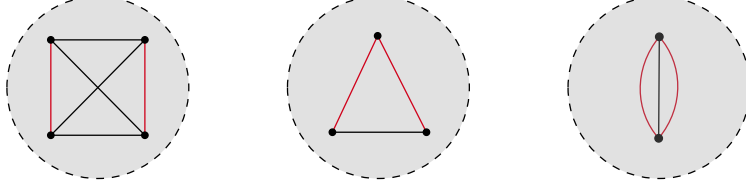


Figure 2: Building blocks of templates

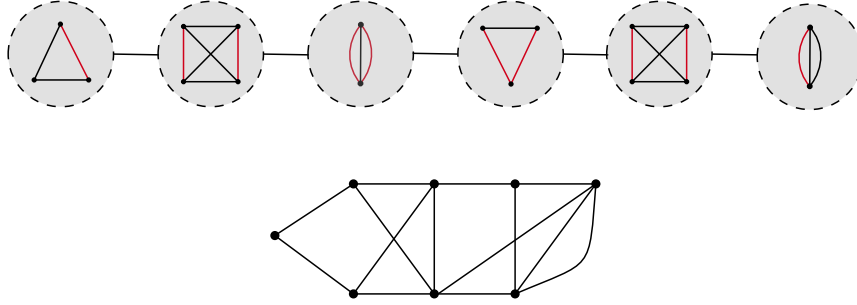


Figure 3: A sample 6-template

2. A fan-type graph F , that is not D_2 , on n spokes is an r -template such that $r \geq n - 1$.
3. For $n \geq 3$, a dipole graph D_n is an $(n - 2)$ -template for which every building blocks is a D_3 .
4. For $n \geq 3$, a cycle C_n is a $(n - 2)$ -template for which every building blocks is a K_3 .

The following is the main theorem of this paper.

Theorem 2.1. *Let r be a positive integer. There exists a function $f_{2.1}(r)$ such that every 2-connected, 3-edge-connected graph G satisfying $|E(G)| \geq f_{2.1}(r)$ contains a parallel extension of an r -template as cc-minor.*

The reason we insist the graph being 3-edge-connected is because we care about the matroid extension of this theorem. Given two matroids M and N , we write $N \leq_{cc} M$ and say that N is a *circuit-contraction minor* or a *cc-minor* of M if there is a sequence $M_0, M_1, \dots, M_k$ of matroids such that $(M_0, M_k) = (M, N)$ and, for all i in $[k]$, there is a circuit C_{i-1} of M_{i-1} such that $M_i = M_{i-1}/C_{i-1}$. Since circuits are the complement of cohyperplanes in matroids (see [6, Proposition 2.1.6]), N is a cc-minor of M if and only

if N^* is a flat restriction of M^* . Thus, we obtain the following matroid corollary:

Corollary 2.2. *Let r be a positive integer. There exists a function $f_{2.2}(r)$ such that every simple, connected, cographic matroid M satisfying $|E(M)| \geq f_{2.2}(r)$ contains a flat N that is isomorphic to $M^*(G)$ where G is a parallel extension of an r -template.*

This corollary is a cographic matroid analogue of the following theorem:

Theorem 2.3 ([1]). *Let r be an integer exceeding two. There is an integer $f(r)$ such that every 2-connected graph of order at least $f(r)$ conduces one of the following: K_r , a clean ladder of order at least r , a member of $\mathcal{K}_{2,r}$, and a member of $\mathcal{K}_{2,r}^+$.*

3 Motivation of cc-minor

Given a graph G , we say H is an *induced subgraph* of G if $H = G \setminus A$ for some $A \subseteq V(G)$. However, in matroid theory there is no natural analogue of vertices. The matroid version of “induced subgraph” is sometimes considered to be “flat”. Let M be a matroid, F is a *flat* of M if $F = cl(F)$ (see [6, Section 1.4]). For graphs, we say H is a *flat* of G if H can be obtained from G by repeatedly deleting edge-cuts and isolated vertices. Clearly, if H is an induced subgraph of G , then H is also a flat of G . However, the other direction does not hold. For example, the disjoint union of two K_2 is a flat of K_4 but not an induced subgraph of K_4 . The following propositions are straight forward if the reader is familiar with graphic matroids.

Proposition 3.1. *Let G be a graph and H a flat of G , then $M(H)$ is a flat of $M(G)$. Particularly, when H is an induced subgraph of G , $M(H)$ is a flat of $M(G)$.*

Recall our definition of circuit-contraction minor. We say a matroid N (or a graph H respectively) is a *cc-minor* of M (or G respectively) if the former can be obtained from the later by repeatedly contracting circuits (or cycles). Knowing that restricting to a hyperplane is same as deleting a cocircuit, the following proposition tells us the relationship between flats and cc-minors.

Proposition 3.2. *Let M be a matroid. N is a flat of M if and only if N^* is a cc-minor of M^* .*

In this paper, we aim to answer the following question:

Question 1. *When M is a sufficiently large ($|E(M)|$ arbitrarily large) simple connected cographic matroid, what are the unavoidable flats of M ?*

We turned this question into its dual version and answer it using graph theory techniques.

Question 2. *When G is a 2-connected, 3-edge-connected graph such that $|E(G)|$ is arbitrarily large, what are the unavoidable cc-minors of G ?*

4 Internally disjoint XY -paths

Let G be a graph and $X, Y \subseteq V(G)$. An XY -path P is an xy -path such that $V(P) \cap X = \{x\}$ and $V(P) \cap Y = \{y\}$. When H and K are subgraphs of G , a $V(H)V(K)$ -path will sometimes be called an HK -path. Two XY -paths P_1, P_2 are internally disjoint if $(V(P_1) \cap V(P_2)) - X - Y = \emptyset$. An X -cycle C is a cycle in G such that $V(C) \cap X \neq \emptyset$.

Lemma 4.1. *Let G be a graph and two disjoint sets $X, Y \subseteq V(G)$. For any X -cycle C such that $V(C) \cap V(Y) = \emptyset$, let $X' = V((G[X] \cup C)/C)$. If there are k internally disjoint XY -paths in G , then there are at least k internally disjoint $X'Y$ -paths in G/C .*

Proof. Let $P_1, P_2, \dots, P_k$ be k internally disjoint XY -paths in G . Let C be an X -cycle not intersecting Y . Contracting C can be viewed as repeatedly contracting edges of C . It suffices to show the following.

4.1.1. *If $G' = G/e$ where e joins u and v and $\{u, v\} \cap X \neq \emptyset$ while $\{u, v\} \cap Y = \emptyset$, then G' contains k internally disjoint $X'Y$ -paths where $X' = V(G[X \cup \{u, v\}]/e)$.*

If $\{u, v\} \subseteq X$ or if $u \in X$ and $v \notin V(P_i)$ for all $i \in [k]$, then $P_1, P_2, \dots, P_k$ are k internally disjoint $X'Y$ -paths in G' . We may now suppose that $u \in X$ and $v \in V(\bigcup_{i \in [k]} P_i) - X$. Since $P_1, P_2, \dots, P_k$ are internally disjoint and $v \notin Y$, there is a unique P_i containing v . We may assume that $v \in V(P_1)$. Let y be the unique vertex in $Y \cap P_1$ and let P'_1 be the yv -path that is a subgraph of P_1 . Now $P'_1, P_2, \dots, P_k$ are k internally disjoint $X'Y$ -paths in G' . Hence 4.1.1 holds. $\square$

Corollary 4.2. *Let G be a k -edge-connected graph and H a cc-minor of G on at least two vertices. Then H is k -edge-connected.*

Proof. Suppose H has a k' -edge-cut and $k' < k$, then there is a partition of $V(H)$ into (X, Y) such that there are k' edges each of them with one end in X and the other end in Y . By Lemma 4.1, we know there must be a partition (X', Y') of G such that there are at most k' edges connecting X' and Y' ; contracting to the fact that G is k -edge-connected. $\square$

Another way to prove Corollary 4.2 is using matroid duality. Note that edge-cuts of G are circuits of $M^*(G)$. Contractions in G are deletions in $M^*(G)$. Since deletions do not create smaller circuit in $M^*(G)$, contractions do not create smaller edge-cuts in G .

5 cc-minors of 2-connected graphs

In this section, we show the unavoidable cc-minor of 2-connected graphs.

Lemma 5.1. *Let G be a 2-edge-connected graph. Then the single vertex clique K_1 is a cc-minor of G .*

Proof. Since G is 2-edge-connected, it does not have any cut-edge. By Corollary 4.2, each cc-minor of G does not have a cut-edge. Let $G_0 = G$. If G_i has an edge e , there must be a cycle C_i of G_i that contains e . Let $G_{i+1} = G_i/C_i$. There is a sequence $G_0, G_1, \dots, G_k$ such that $E(G_k) = \emptyset$. Therefore, $G_k \cong K_1$. $\square$

Taking the advantage of this lemma, we say contracting a 2-edge-connected graph H of G meaning a sequence of cycle contraction that is equivalent to contracting all the edges of H . The following lemma is a strengthen of the previous one. It tells us the unavoidable cc-minor of 2-connected graphs if we want to preserve a specified edge. A *parallel connection* G of G_1, G_2 is obtained by identifying nonloop edge e_1 of G_1 and e_2 of G_2 and keeping the identified edge, which we call it the *base point*. Similarly as 2-sum, parallel connection depends on the choice of e_1, e_2 , and how e_1, e_2 are identified (how their ends are paired).

Lemma 5.2. *Let G be a 2-edge-connected graph and suppose e is an edge of G joining u and v . Then G has a cc-minor H that is the parallel connection with base point e of a collection of cycles containing e .*

Proof. Let $G_0 = G$. If G_i has a cycle C_i which does not contain both u and v , then let $G_{i+1} = G_i/C_i$. There is a sequence $G_0, G_1, \dots, G_k$ such that every cycle in G_k contains both u and v . We claim that $H = G_k$ is the parallel connection with base point e of a collection of cycles containing e .

Suppose $w \notin \{u, v\}$ is a vertex of H . We first prove that $\deg(w) = 2$. By Lemma 4.2, H is 2-edge-connected, so $\deg(w) \geq 2$. Since every cycle in H containing both u and v , we know w is in a cycle C containing u and v . Suppose $\deg(w) \geq 3$, there is an edge $f \notin C$ joining w and another vertex x . Let $H' = H \setminus f$. Because H is 2-edge-connected, H' is connected. Choose P to be the shortest path in H' from x to C . Let P^+ be the path in H which consists of P and the edge f . We know $E(P^+) \cap E(C) = \emptyset$ and

$V(P^+) \cap V(C) = \{w, y\}$ for some $y \in V(C)$. Since not both u and v are in P^+ , there is a cycle of H does not contain both u and v ; a contradiction.

Let A be the set of degree 2 vertices in H . If we suppress $A \setminus \{u, v\}$, the resulting graph H' will be a dipole graph D_n ($n \geq 2$) on u and v . Hence H is the parallel connection with base point e of a collection of cycles containing e . $\square$

Combining Lemma 4.2 and Lemma 5.2, we get the following:

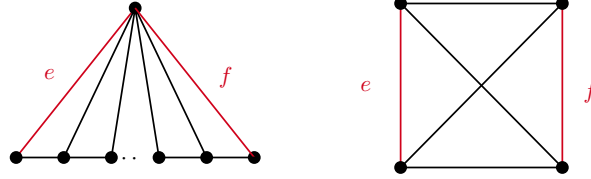
Corollary 5.3. *Let G be a 3-edge-connected graph and suppose e is an edge of G joining u and v . Then G has a cc-minor H that is a dipole graph D_n on u, v containing e . Moreover, $n \geq 3$.*

6 cc-minors of 3-connected graphs

In this section, we show the unavoidable cc-minors of any 3-connected graph. This result helps us to analyze the block decomposition of large 2-connected graphs.

Theorem 6.1. *Let G be a simple 3-connected, 3-edge-connected graph and e, f two distinct edges in G . Then G has a cc-minor H containing e, f such that either one of following is true:*

- i. *G is isomorphic to a fan-type graph and e, f are two outer spokes; or*
- ii. *The simplification of H , denoted by $si(H)$, is isomorphic to K_4 and e, f are two non-adjacent edges.*



Proof. Let e be the edge joining x_1 and x_2 and let f be the edge joining y_1 and y_2 . First we prove the following:

6.1.1. *If e and f are two adjacent edges in G , then G has a cc-minor H such that H is isomorphic to a fan-type graph and e, f are two outer spokes of H .*

If e and f are two parallel edges, then by Lemma 5.2, G has a cc-minor $H \cong D_n$ containing e and f . Note that dipole graphs are special fan-type graphs. Hence 6.1.1 holds. Now, without loss of generality, we may assume $x_1 = y_1$ and $x_2 \neq y_2$. Since G is 3-connected, $G \setminus x_1$ is 2-connected. By Lemma 5.1, there is a sequence $G_0^-, G_1^-, \dots, G_k^-$ such that $(G_0^-, G_k^-) = (G \setminus x_1, K_1)$ and, for all i in $[k]$, there is a cycle C_{i-1} of G_{i-1}^- such that $G_i^- = G_{i-1}^- / C_{i-1}$. Therefore, let $G_0 = G$ and, for all i in $[k]$, let $G_i = G_{i-1}^- / C_{i-1}$. The sequence $G_0, G_1, \dots, G_k$ is a sequence of cc-minors such that $(G_0, G_k) = (G, D_m)$. Moreover, $e, f \in E(G_k)$. Hence 6.1.1 holds.

Now we may assume e and f are two non-adjacent edges. For any two vertices u and v , a θ -graph on (u, v) consists of three internally disjoint uv -paths. For any $(x, y) \in \{x_1, x_2\} \times \{y_1, y_2\}$, a θ -graph on (x, y) containing both e and f is *type-A* if there is an xy -path containing both e and f , and is *type-B* if e and f belongs to two different xy -path. We then prove the following:

6.1.2. *For each pair $(x, y) \in \{x_1, x_2\} \times \{y_1, y_2\}$, there is a θ -graph Θ on (x, y) such that $\{e, f\} \subset E(\Theta)$.*

Without loss of generality, we assume $x = x_1$ and $y = y_1$. By Menger's theorem, we know there three internally disjoint x_1y_1 -paths form a θ -graph. We call this graph Θ_0 . If $\{e, f\} \subset E(\Theta_0)$, then 6.1.2 holds. So we may assume $e \notin E(\Theta_0)$. Because $G \setminus x_1$ is connected, there is a x_1y_1 -path $P = p_0p_1 \dots p_k$ where $p_0 = x_1$, $p_1 = x_2$ and $p_k = y_1$. Let $i \geq 2$ be the minimal index such that $p_i \in V(\Theta_0)$. If $p_i \neq y_1$, then we know p_i belongs to a x_1y_1 -path Q of Θ_0 . Then we can replace the x_1p_i -path of Q by the x_1p_1 -path of P and call the new θ -graph Θ_1 . Otherwise, we know $p_i = y_1$. Pick Q to be an x_1y_1 -path in Θ_0 such that $f \notin E(Q)$. Let Θ_1 be the θ -graph obtained from Θ_0 by replacing Q by P . If $f \in E(\Theta_1)$, let $\Theta = \Theta_1$. Otherwise, we repeat this process. Hence, 6.1.2 holds.



Figure 4: Type-A θ -graph Θ_A and type-B θ -graph Θ_B

We first prove that a type-A θ -graph is a winning case for us.

6.1.3. *If G has a cc-minor G' that has a type-A θ -graph Θ_A as subgraph, then G has a cc-minor H such that H is isomorphic to a fan-type graph and e, f are two outer spokes of H .*

Since $\leq_{cc}$ is transitive, and by Lemma 4.2 any cc-minor of a 3-edge-connected graph is 3-edge-connected. It suffices to show that if a 3-edge-connected graph G has type-A θ -graph Θ_A as subgraph, then G has a cc-minor H such that H is isomorphic to a fan-type graph and e, f are two outer spokes of H .

Observe that there is a cycle in Θ_A not containing e and f . We call this cycle C . Let $G' = G/C$ and we call the identified vertex x . Since e and f are contained in a cycle of G' , there is a 2-block B of G' containing both e and f . Since G' does not have any cut-edge, by Lemma 5.1, we may contract all the 2-blocks of G' except B . Note that the resulting graph is isomorphic to B , so we just call it B . Let $B_0 = B$ and $B_i^- = B_i \setminus x$. If B_i^- has a cycle C_i , let $B_{i+1} = B_i/C_i$. There is a sequence $B_0, B_1, \dots, B_k$ such that B_k^- is a tree and we call this tree T . Let P be the x_2y_2 -path in T (it may be a single vertex if x_2 and y_2 are identified). Let $H_0 = B_k$ and $T_i = H_i \setminus x$. If $T_i \neq P$, then there is a leaf $l_i \in V(T_i) - V(P)$. By Lemma 4.2, $\deg(l_i) \geq 3$ so there are at least two xl_i -edges forming a cycle O_i . Let $H_{i+1} = H_i/O_i$. There is a sequence $H_0, H_1, \dots, H_k$ such that $H_k \setminus x = P$. Since each vertex on P has degree at least 3, there is an xp -edge for all $p \in V(P)$. Hence $H = H_k$ is a fan-type graph and e, f are two outer spokes of H , and 6.1.3 holds.

Now we may assume G has a type-B θ -graph Θ_B as subgraph. Without loss of generality, we suppose Θ_B is on (x_1, y_1) . By abuse of notation, for any type-B θ -graph on (x_1, y_1) . We call the x_2y_1 -path P_1 , the x_1y_1 -path P_2 , and the x_1y_2 -path P_3 as shown in Figure 5.

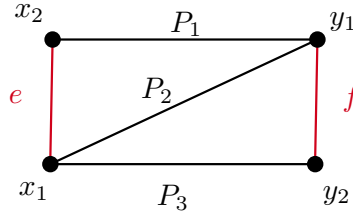
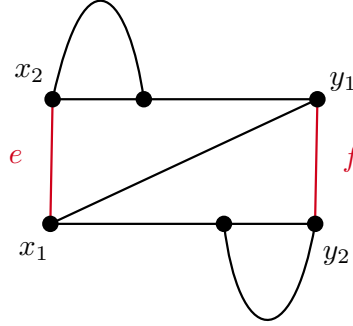


Figure 5: Θ_B

Let $(G_0, \Theta_0) = (G, \Theta_B)$. If G_i has a path $P_i = p_0^i p_2^i \dots p_k^i$ such that $p_0^i = x_2$ ($p_0^i = y_2$ respectively), $p_k^i \in V(P_1) - y_1$ ($p_k^i \in V(P_3) - x_1$ respectively), and internally disjoint with Θ_i . then let C_i be the cycle formed by P and

the x_2p_k -path on P_1 (the y_2p_k -path on P_3 respectively). Let $(G_{i+1}, \Theta_{i+1}) = (G_i/C_i, \Theta_i/C_i)$. There is a sequence $(G_0, \Theta_0), (G_1, \Theta_1), \dots, (G_t, \Theta_t)$ such that in G_t there is no such path satisfying the condition. We may assume P_1 and P_3 each has at least one edge, otherwise by 6.1.3 the theorem holds.



Let $\mathcal{P}_x$ be the collection of path starting at x_2 , end on $V(\Theta_t)$ and internally disjoint with Θ_t . Let $\mathcal{P}_y$ be the collection of path starting at y_2 , end on $V(\Theta_t)$ and internally disjoint with Θ_t . If there is a path $P \in \mathcal{P}_x$ which ends on $V(P_2) - x_1$ then there is a type-A θ -graph on (x_2, y_1) (see Figure 6), and by 6.1.3 the theorem holds.

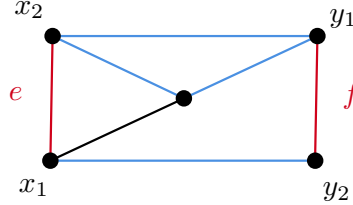


Figure 6: type-A θ -graph on (x_2, y_1) in G_t

If all the path in $\mathcal{P}_x$ ends on x_1 , then $N_{\Theta_t}(x_2)$ is a 2-vertex-cut in G ; contracting to G being 3-connected. So we may assume there is a path $P_x \in \mathcal{P}_x$ which ends on $V(P_3) - x_1$. By symmetry, we may also assume there is a path $P_y \in \mathcal{P}_y$ which ends on $V(P_1) - x_1$. If P_x and P_y are internally disjoint, then there is a type-A θ -graph on (x_2, y_2) (see Figure 7), and by 6.1.3 the theorem holds.

Now we may assume P_x and P_y are not internally disjoint. Therefore there is a x_2y_2 -path P_4 in t which is internally disjoint with Θ_t . Let K be

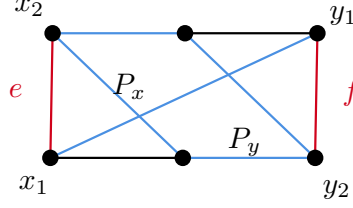


Figure 7: type-A θ -graph on (x_2, y_2) in G_t

the subgraph in G_t consisting of Θ_t and P_4 (see Figure 8).

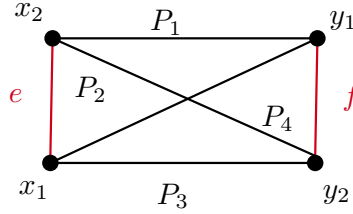


Figure 8: Subgraph K in G_t

By Menger's theorem and Lemma 4.1, we know for each $v \in V(G_t) - V(K)$ there are at least three internally disjoint vK -paths. If there are two such paths lands on $\{a, b\} \subseteq V(P_i)$ for $i \in [4]$, let C_v be the cycle formed by the va -path, the ab -path, and the bv -path. Note that if $\{a, b\} \in \{x_1, x_2\} \times \{y_1, y_2\}$, by 6.1.3 the theorem holds. So we may assume contracting C_v will not make any of P_i become a single vertex. Now suppose no single P_i receives two ends of vK -paths. For $i \in [4]$, let P_i^- be the path P_i deleting two end vertices. Observe that there must be two internally disjoint vK -path with ends a, b such that $a \in P_i^-$ and $b \in P_j^-$ for $(i, j) \in \{(1, 2), (1, 4), (2, 3), (2, 4)\}$. Without loss of generality, we assume $a \in P_1^-$ and $b \in P_4^-$. Let C_v be the cycle formed by the va -path, the ay_1 -path, the y_1b -path, and the bv -path. Since $\{a, b\} \cap \{x_1, x_2, y_1, y_2\} = \emptyset$, contracting C_v will not make any of P_i become a single vertex. Let $(F_0, K_0) = (G_t, K)$. If $v \in V(F_i) - V(K_i)$, let $(F_{i+1}, K_{i+1}) = (F_i/C_v, K_i/C_v)$. We repeat this process until $V(F_m) = V(K_m)$. By abuse of notation, let $F = F_m$ and $K = K_m$.

Let $(H_0, J_0) = (F, K)$. If there exists $u \in V(H_i) - \{x_1, x_2, x_3, x_4\}$. Since $\deg(u) \geq 3$, u is incident to an edge $uw \notin E(J_i)$. Then we discuss three cases (see Figure 10).

Case 1: (u and w belongs to same P_i for $i \in [4]$) Let C_u be the cycle formed by edge uw and the wu -path on P_i . Let $(H_{i+1}, J_{i+1}) =$

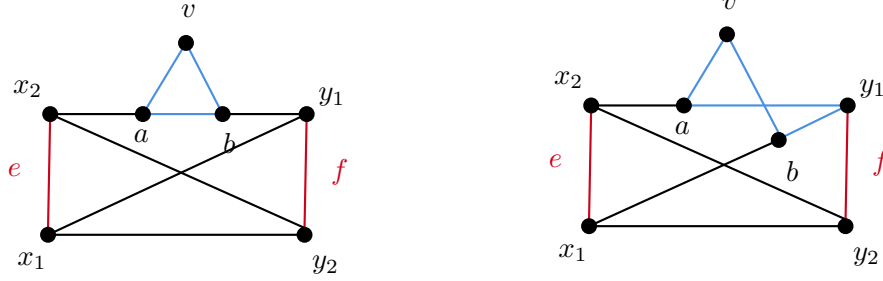


Figure 9: Absorbing an external vertex v into Θ_B

$(H_i/C_u, J_i/C_u)$.

Case 2: ($u \in V(P_i)$, $w \in V(P_j)$, and $V(P_i) \cap V(P_j) = z$) Let C_u be the cycle formed by edge uw , the wz -path on P_j and the zu -path on P_i . Let $(H_{i+1}, J_{i+1}) = (H_i/C_u, J_i/C_u)$.

Case 3: ($u \in V(P_i)$, $w \in V(P_j)$, and $V(P_i) \cap V(P_j) = \emptyset$) Then F_i has a type-A θ -graph, and by 6.1.3 the theorem holds.

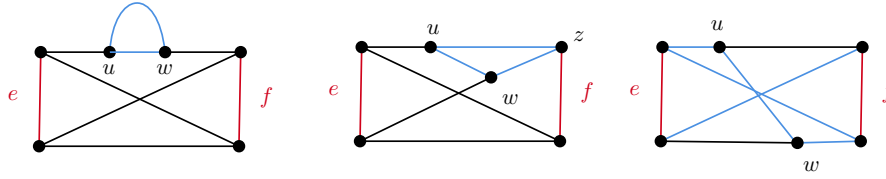


Figure 10: Three cases of uw -edge

Therefore, there is a sequence $(H_0, J_0), (H_1, J_1), \dots, (H_n, J_n)$. Let $H = H_n$. By our construction $V(H) = \{x_1, x_2, y_1, y_2\}$, $\{e, f\} \subset E(H)$, and $si(H) \cong K_4$.

□

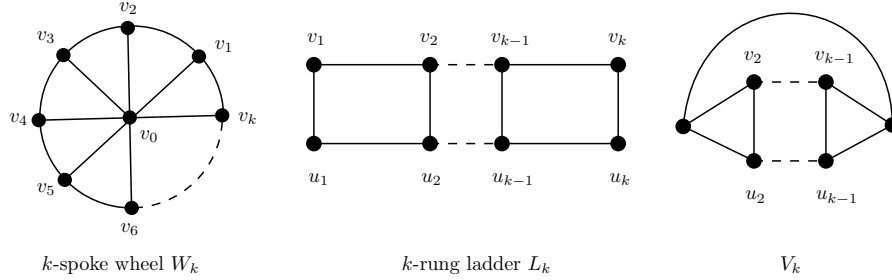
7 cc-minors of large 3-connected graphs

In this section, we show the unavoidable cc-minors of sufficiently large 3-connected graphs.

Theorem 7.1. *For every positive integer t , there is a function $f_{7.1}(t)$ such that if a simple, 3-connected, 3-edge-connected graph G has at least $f_{7.1}(t)$ edges, then G has a fan-type graph $F_{t_1, t_2, \dots, t_n}$ as a circuit-contraction minor such that $\sum_{i=1}^n t_i \geq t$.*

Before beginning the proof of Theorem 7.1, we present two more lemmas. The first is a Ramsey-type result for 3-connected graphs and the second is a result for weighted trees, which we used as auxiliary graphs in our analysis.

The following lemma by Oporowski, Oxley, and Thomas (see [5] or [4, Proposition 9.4.3]) characterizes the unavoidable structures of large 3-connected graphs. The k -spoke wheel, denoted by W_k , has vertices $v_0, v_1, v_2, \dots, v_k$, where $v_1, v_2, \dots, v_k$ form a cycle, and v_0 is adjacent to all of $v_1, v_2, \dots, v_k$. The k -rung ladder, denoted by L_k , has vertices $v_1, v_2, \dots, v_k, u_1, u_2, \dots, u_k$, where $v_1, v_2, \dots, v_k$, and $u_1, u_2, \dots, u_k$ form paths in the listed order, and v_i is adjacent to u_i for $i = 1, 2, \dots, k$. The graph V_k is obtained from L_k by adding an edge between v_1 and v_k , and contracting the edges joining u_1 to v_1 and u_k to v_k .



Lemma 7.2. *For every integer $k \geq 3$, there is a function $f_{7.2}(k)$ such that every 3-connected graph with at least N vertices contains a subgraph isomorphic to a subdivision of one of W_k , V_k , and $K_{3,k}$.*

A *weighted tree* is a tree T together with a weight function $w : 2^{V(T)} \rightarrow \mathbb{Z}^+$ such that each vertex v is given a weight $w(v)$ and for each $A \subseteq V(T)$, $w(A) = \sum_{v \in A} w(v)$. We prove the following lemma for weighted trees.

Lemma 7.3. *For any integer t exceeding one, there is a function $f_{7.3}(t)$ such that every weighted tree T satisfying $w(V(T)) \geq f_{7.3}(t)$ contains one of the following:*

1. a vertex v such that $\deg(v) > t$;
2. a path P such that $|V(P)| > t$;

3. a path P such that $w(V(P)) > t$.

Proof. We prove that $f_{7.3}(t) = \sum_{i=1}^{\lceil \frac{t}{2} \rceil} t^i$ satisfies the condition. Suppose T is a tree such that both the maximum degree and diameter of T do not exceed t , then $|V(T)| < \sum_{i=0}^{\lfloor \frac{t}{2} \rfloor} t^i$. However, since $w(T) \geq \sum_{i=1}^{\lceil \frac{t}{2} \rceil} t^i$, there exists a vertex v such that $w(v) > t$. Thus any path P containing v satisfies $w(V(P)) > t$. $\square$

We are now ready to prove the main result of this section.

Proof of Theorem 7.1.

Let H be a subgraph of G . We say $e \in E(G) - E(H)$ is an H -bridge if e is incident with at least one vertex of H . We call each connected component of $G \setminus V(H)$ a H -island. Note that there are two types of H -bridges:

1. an edge consists of two vertices of H ;
2. an edge consists of one vertex of H and one vertex of a H -island.

We first prove the following:

7.3.1. *If G has a cycle C and a C -island I such that there are at least $f_{7.3}(t)$ C -bridges between C and I , then G has a fan-type graph $F_{t_1, t_2, \dots, t_n}$ as a cc-minor such that $\sum_{i=1}^n t_i \geq t$.*

First observe that $G \setminus V(I)$ does not have a cut-edge. By Lemma 5.1, K_1 is a cc-minor of $G \setminus V(I)$. After contracting all the cycles in $G \setminus V(I)$, the resulting graph, denoted by G' , is obtained from $G[V(C) \cup V(I)]$ by contracting C . Note that G has a vertex c such that $\deg(c) \geq f_{7.3}(t)$ and $N(c) \subseteq V(I)$. Let $G_0 = G'$. If $G_i \setminus c$ has a cycle C_i , let $G_{i+1} = G_i / C_i$. There is a sequence $G_0, G_1, \dots, G_k$ such that $G_k \setminus c = T$ is a tree. Now define the weight function $w : V(T) \rightarrow \mathbb{Z}^+$. For each vertex v of T , $w(v)$ is the number of edges joining c and v . Clearly, $w(V(T)) = \deg(c) \geq f_{7.3}(t)$. By Lemma 7.3, T has a subgraph T' which is one of the following:

1. a vertex v such that $\deg(v) > t$;
2. a path P such that $|V(P)| > t$;
3. a path P such that $w(V(P)) > t$.

Let $(H_0, T_0) = (G_k, T)$. If $T_i \neq T'$, then T_i has a leaf $l \notin V(T')$. Since H_i is 3-edge-connected, there are two edges joining c and l that form a cycle O_i in H_i . Let $H_{i+1} = H_i / O_i$ and $T_{i+1} = T \setminus l$. Thus, there is a sequence

$(H_0, T_0), (H_1, T_1), \dots, (H_k, T_k)$ such that $T_k = T'$. By the choice of T' , H_k is a fan-type graph which has at least t spokes. Thus, 7.3.1 holds.

Now we claim $f_{7.1}(t) = \binom{f_{7.2} \circ f_{7.3}(t)}{2}$ satisfies the condition.

Since G is simple and has more than $\binom{f_{7.2} \circ f_{7.3}(t)}{2}$ edges, G has more than $f_{7.2} \circ f_{7.3}(t)$ vertices. By Lemma 7.2, G has a subgraph isomorphic to a subdivision of one of W_k , V_k , and $K_{3,k}$ where $k \geq f_{7.3}(t)$. For each of three cases, let C be the red cycle and I be the C -island containing the blue vertices as shown in Figure 11. It is straight forward to check that the choices of C and I satisfying the conditions in 7.3.1. Hence by 7.3.1, G has a fan-type graph $F_{t_1, t_2, \dots, t_n}$ as a cc-minor such that $\sum_{i=1}^n t_i \geq t$. $\square$

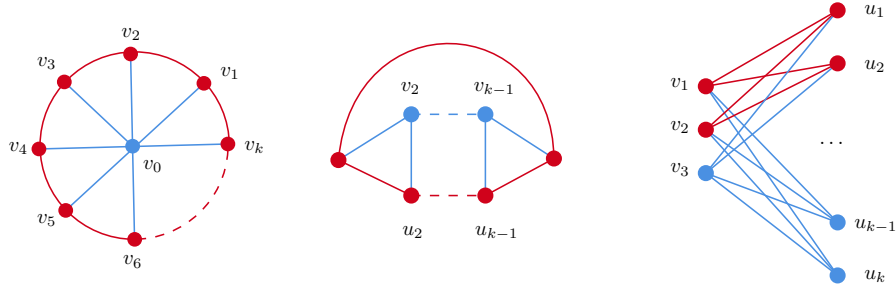


Figure 11: C and C -island I in W_k , V_k , and $K_{3,k}$

8 Proof of Theorem 2.1

The proof of our main theorem uses the decomposition lemma of 2-connected graphs by Tutte [10].

Lemma 8.1. *Every 2-connected graph G can be constructed from 3-connected minors of G , K_3 , and D_3 by taking 2-sums.*

Let G be a 2-connected graph. Taking the advantage of Lemma 8.1, we can display a *tree-decomposition* of G by a tree T (see Figure 12), where each vertex v of T corresponds to a decomposed basic graph G_v (G_v is isomorphic to a 3-connected minor of G , K_3 , or D_3) and each edge uv of T corresponds to the common edge of G_u and G_v (over which a 2-sum takes place). In this sense, we can also say that G is a *tree-structure* of 3-connected graphs, K_3 , and D_3 .

Now we are ready to present the proof of our main theorem.

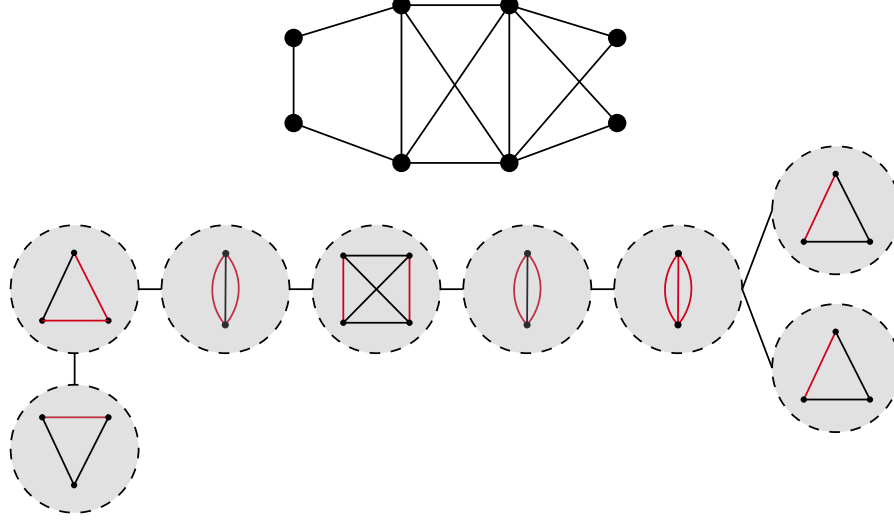


Figure 12: A 2-connected graph G and its tree-decomposition

Proof of Theorem 2.1.

Let $f_{7.1}(r+1) = R$. We claim that $f_{2.1}(r) = r \cdot \sum_{i=1}^{\lfloor \frac{r}{2} \rfloor + 1} R^i$ satisfy the condition.

Let G be a 2-connected, 3-edge-connected graph such that $|E(G)| \geq (r+2) \cdot \sum_{i=1}^{\lfloor \frac{r}{2} \rfloor + 1} R^i$.

If G has a parallel class $\mathcal{P}$ such that $|\mathcal{P}| \geq (r+2)$, then by Lemma 5.2 G has a dipole graph D_n as cc-minor and $n \geq r+2$. Recall that a dipole graph D_n is a $(n-2)$ -template. Thus, the theorem holds.

So we may assume $si(G)$ is a 2-connected, 3-edge-connected graph such that $|E(si(G))| \geq \sum_{i=1}^{\lfloor \frac{r}{2} \rfloor + 1} R^i$. Let T be tree-decomposition of G based on its tree-structure of 3-connected graphs, K_3 , and D_3 . If there is a vertex $v \in V(T)$ such that its corresponding decomposed basic graph G_v has at least R edges. By Lemma 7.1, we know G_v has a cc-minor which is a fan-type graph F with at least $r+1$ spokes. By Lemma 5.2, G has a cc-minor F' which is a parallel extension of F . Recall that a fan-type graph on $r+1$ spokes is an m -spoke such that $m \geq r$. Hence, the theorem holds.

Now we may assume, for all $v \in V(T)$ and its corresponding G_v , $|E(G_v)| \geq R$. Therefore, T either has a vertex v such that $\deg(v) \geq R$ or a path P such that $|V(P)| \geq r$. However, since $|E(G_v)| < R$ for all v , we cannot apply 2-sum R times on any decomposed basic graph. So T does not have a vertex of degree at least R .

Let $P = p_0 p_1 \dots p_k$ be the path in T such that $|V(P)| \geq r$. We know for each p_i , its corresponding decomposed basic graph is one of 3-connected minor of G , K_3 , or D_3 . Moreover, $p_i p_{i+1}$ corresponds to the edge of G_{p_i} and $G_{p_{i+1}}$ over which a 2-sum takes place. By Lemma 5.2, G_{p_0} and G_{p_k} has cc-minor G_0 and G_k such that G_0 is a dipole graph containing $p_0 p_1$ and G_k is a dipole graph containing $p_{k-1} p_k$. For $0 < i < k$, if G_{p_i} is one of K_3 or D_3 , then let $G_i = G_{p_i}$. Otherwise, by Lemma 6.1, G_{p_i} has a cc-minor G_i such that G_i is either a fan-type graph where $p_{i-1} p_i$ and $p_i p_{i+1}$ are two outer spokes or a parallel extension of K_4 where $p_{i-1} p_i$ and $p_i p_{i+1}$ are two non-adjacent edges. Thus, we know $G_{p_0} \oplus_2 G_{p_1} \oplus_2 \dots \oplus_2 G_{p_k}$ has a cc-minor $G_0 \oplus_2 G_1 \oplus_2 \dots \oplus_2 G_k$ which is a k -template where $k \geq r$. Then, by Lemma 5.2, we know G has a cc-minor H which is a parallel-extension of k -template. Hence, the theorem holds. $\square$

References

- [1] S. Allred, G. Ding, B. Oporowski, Unavoidable induced subgraphs of large 2-connected graphs, (2020) [*arXiv:2009.12503*](#)
- [2] Y. Chen, T. Mansour and Q. Zou, Embedding distributions of generalized fan graphs, *Canad. Math. Bull.* **56** (2013), 265–271.
- [3] W. H. Cunningham and J. Edmonds, A combinatorial decomposition theory, *Canad. J. Math.* **32** (1980), 734–765.
- [4] R. Diestel, *Graph Theory* (Fifth Edition) volume 173, Graduate Texts in Mathematics. Springer, Berlin (2018).
- [5] B. Oporowski, J. Oxley, and R. Thomas Typical subgraphs of 3- and 4-connected graphs, *J. Combin. Theory Ser. B* **57** (1993), 239–257.
- [6] J. Oxley, *Matroid Theory* (Second Edition), Oxford Graduate Texts in Mathematics. Oxford University Press, Oxford (2011).
- [7] F. P. Ramsey, On a problem of formal logic, *Proc. London Math. Soc.* **30** (1930), 264–286.
- [8] P. D. Seymour, Decomposition of regular matroids, *J. Combin. Theory Ser. B* **28** (1980), 305–359.
- [9] P. D. Seymour, On minors of non-binary matroids, *Combinatorica* **1** (1981), 387–394.
- [10] W. T. Tutte, *Connectivity in graphs*. University of Toronto Press, Toronto (1966).
- [11] W. T. Tutte, *Graph Theory*, Encyclopedia of Mathematics and its application. Cambridge University Press, Cambridge (2001).